ST.JOSEPH'S
COLLEGE OF ENGINEERING
AND TECHNOLOGY,
- PALAI -
AUTONOMOUS

Department of
Electronics &
Computer
Engineering

er

MULTIPURPOSE AUTONOMOUS AND PILOTED AIRCRAFT SYSTEM (M.A.P.A.S)

BATCH No:13

TEAM MEMBERS:
Edwin Baiju(SJC22ECE024)
George Siby(SJC22ECE027)
Govind P (SJC22ECE030)
Ron Thomas(SJC22ECE053)

GUIDE:
Mr Al-Saheer SS
ASST.PROFESSOR
DEPT. OF ELECTRONICS & COMPUTER ENGINEERING

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- **Problem statement:**

Traditional drones are limited by single-purpose designs, often compromising endurance, payload capacity, or precision. This lack of flexibility results in inefficiencies and increased costs for missions that require both wide-area coverage and detailed close-range operations.

- **Mission statement:**

To design and develop a modular, multi-purpose autonomous drone system capable of performing diverse mission-critical tasks through intelligent software control and interchangeable payload attachments, thereby improving operational flexibility, efficiency, and response time in real-world scenarios.

- **Target users/customers**

- Government agencies can use it for disaster rescue, surveillance, and emergency announcements.
- Farmers benefit from precise spraying and crop monitoring.
- NGOs and relief teams can deliver food and medicine to remote areas.
- Businesses can use it for fast logistics and medical supply delivery.

- **Social relevance/use/importance:**

. Reduces human risk in dangerous disaster zones.

. Supports farmers with safer and efficient spraying methods.

. Enables contactless delivery during pandemics or emergencies.

. One modular drone replaces multiple single-purpose drones, reducing cost and increasing flexibility.

. Promotes innovation in autonomous systems and smart UAV technology for real-world societal benefit.

Literature Review:

TITLE	AUTHOR	YEAR	FINDINGS
A Drone Technology Implementation Approach to Conventional Paddy Fields Application	S.D. Panjaitan, Y.S.K. Dewi, M.I. Hendri, R.A. Wickasono and H. Priyatman	2023	The study found that implementing a hexacopter agricultural drone in conventional paddy fields significantly improved farming efficiency and crop growth. The drone, capable of carrying 16 liters of fertilizer or pesticide, reduced spraying time to less than one-third of the traditional method while minimizing farmers' exposure to toxic chemicals.
FPV Drone with GPS used for Surveillance in Remote Areas	Apurv Saha, Akash Kumar, Aishwary Kumar Sahu	2023	The paper presents a low-cost FPV (first-person-view) quadcopter system with onboard GPS aimed at surveillance in hard-to-reach areas. The design integrates real-time video (with audio) transmission to an operator, GPS-based geo-tagging/position feedback, and multiple operating modes to extend range and endurance.

TITLE	AUTHOR	YEAR	FINDINGS
Drone-Assisted Public Safety Networks: The Security Aspect	Daojing He, Sammy Chan, and Mohsen Guizani	2023	This paper examines how drones can enhance public safety networks by providing mobile communication support in disaster or emergency situations. While drones help fill coverage gaps and enable faster deployment, their use introduces significant cybersecurity risks such as GPS spoofing, Wi-Fi hacking, denial-of-service, and data interception. The study stresses the need for strong security measures—encryption, authentication, and intrusion detection—to ensure confidentiality, integrity, and availability of critical communications
Drone Food Delivery: A Solution to Crowding During the Global COVID-19 Pandemic	Gurmeet Singh , Shavneet Sharma , Anushree Tandon, and Puneet Kaur	2025	This reference discusses the role of drones in disaster management and humanitarian operations. It highlights drones' ability to perform rapid aerial surveillance, deliver emergency supplies, and assist in search-and-rescue missions. Findings suggest that drones improve efficiency, safety, and response time compared to traditional methods, especially in areas inaccessible to humans or ground vehicles.

TITLE	AUTHOR	YEAR	FINDINGS
Design of the Life-ring Drone Delivery System for Rip Current Rescue	Gang Xiang, Andrew Hardy, Mohammed Rajeh, Lahari Venuthurupalli	2022	This paper presents the design of a drone-based life-saving system for rip current rescues on beaches. It shows that an octocopter carrying a tethered life-ring can reduce rescue time by 39% and significantly increase survival probability for drowning victims compared to lifeguards alone. Simulation and system modeling determined that weight capacity and battery power are the main design constraints.

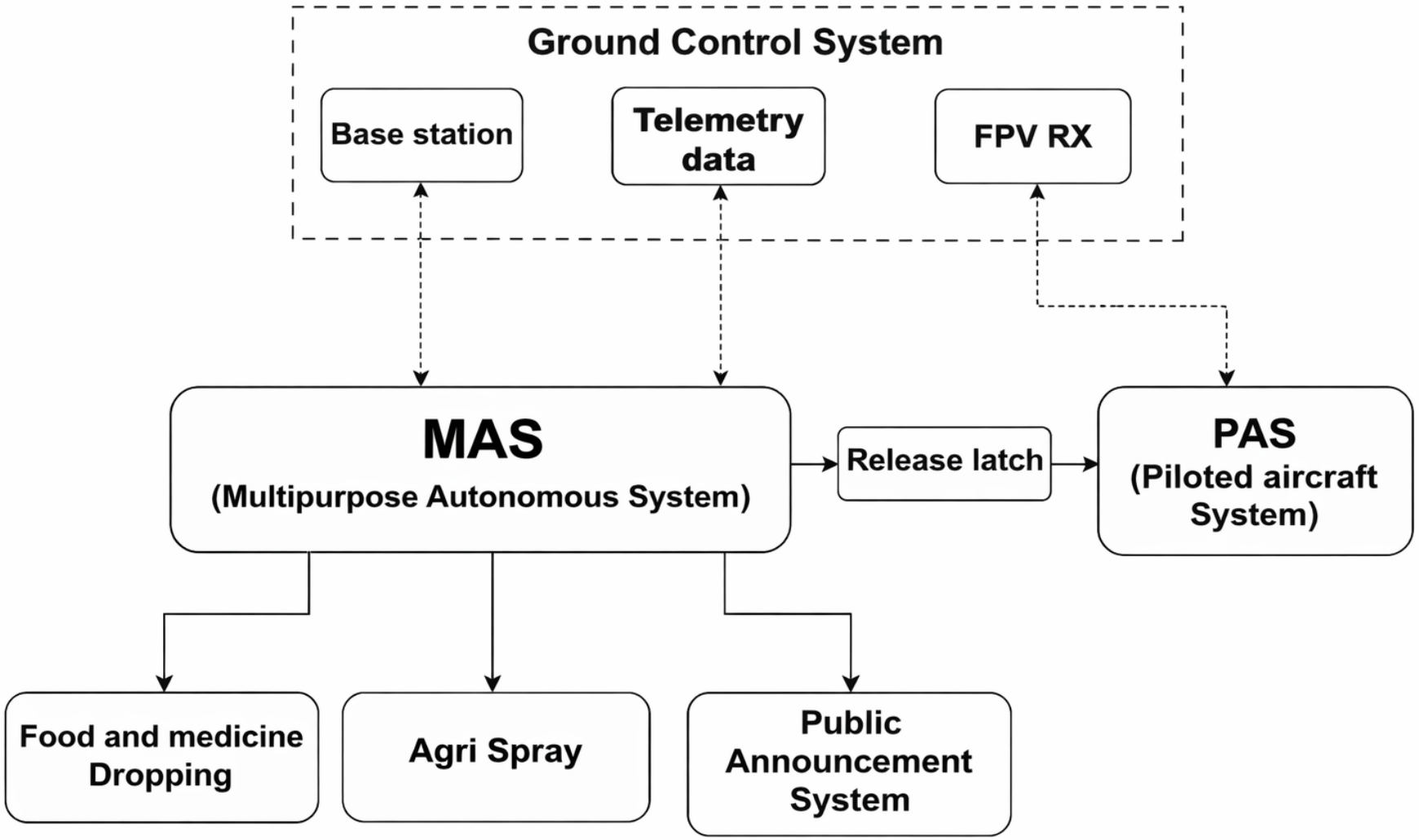
Field Survey

GPS Map Camera
Manarcadu, Kerala, India
Hhxp+v2p, Manarkadu Church Jn Thiruvanchur Rd, Manarcadu, Kerala 686019, India
Lat 9.599978° Long 76.58541°
05/09/2025 11:07 AM GMT +05:30
Google

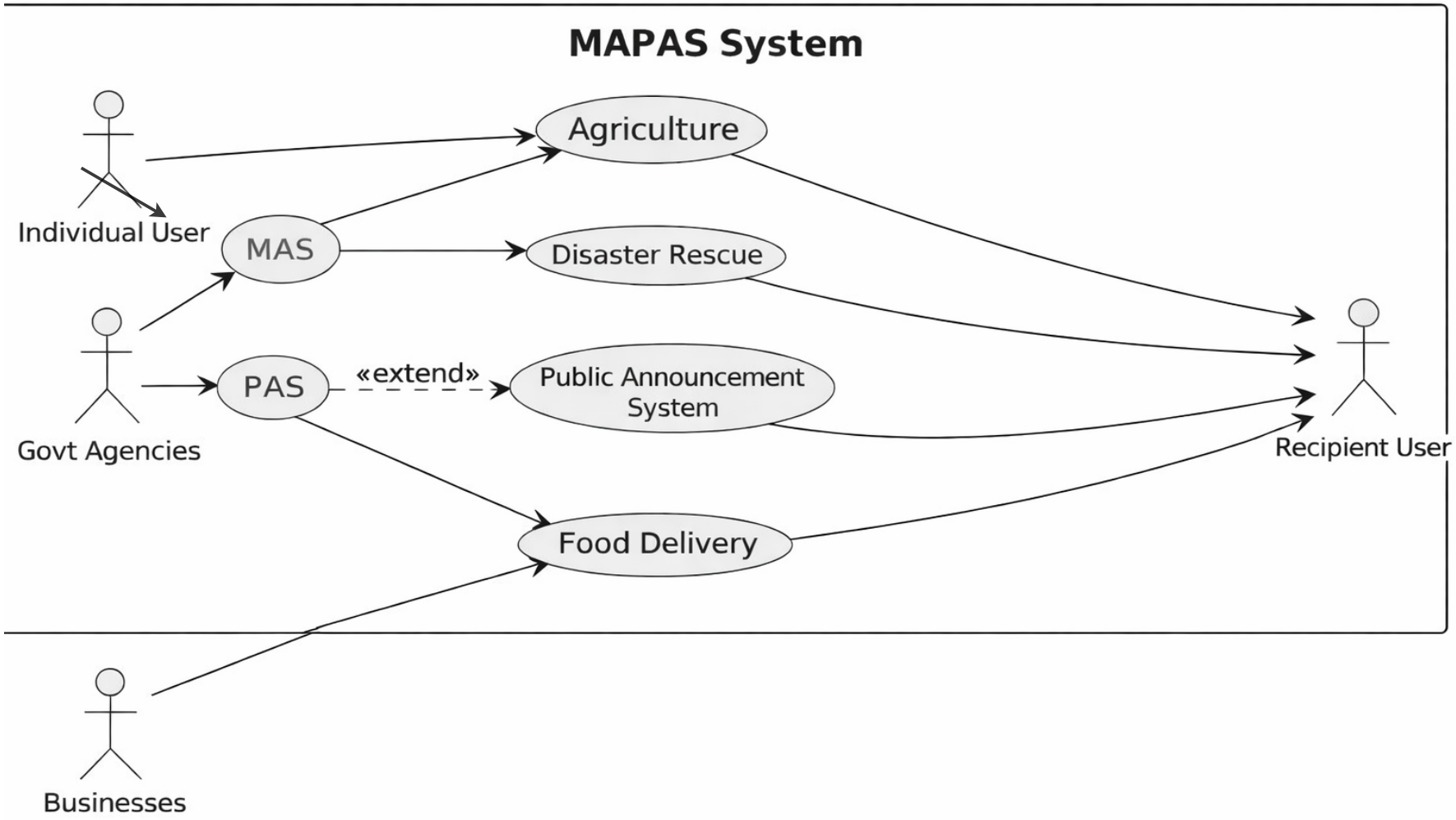
Need Metrics:

	Metrics	1	2	3	4	5	6	7
	Interpreted needs	Payload capacity	Range	Surveillance	Agri efficiency	Safety & Stability	Modularity	Autonomous
1	The drone should deliver life-saving supplies (food, medicine, lifebuoys) in disaster zones.	X		X				X
2	The drone should reach inaccessible/remote areas.		X			X	X	
3	The drone should support real-time communication and surveillance for emergency teams.		X	X			X	
4	The drone should spray pesticides and monitor crops efficiently.			X	X			
5	The drone should operate safely in crowds/events.			X		X		X
6	The drone should be modular and cost-effective.	X			X		X	
7	The drone should allow both autonomous and piloted			X		X		X
8	The drone should withstand harsh conditions (wind, rain, etc.).		X		X	X		

Detailed Block Diagram:



Use-case Diagram:



Software & Hardware Specifications:

Hardware:

<u>MAS</u>

- **Pixhawk 6X:** Flight controller for autonomous navigation.
- **Skydroid T12 Radio System:** Manual control and telemetry link.
- **6S 22000 mAh LiPo:** Provides power to motors and electronics.
- **T-Motor Antigravity Motors:** Propulsion for heavy-lift capabilities.
- **40A ESCs:** Regulate motor speed and power delivery.
- **ZD850mm Frame:** Structural body supporting all components and payloads.

Companion Computer

- **Raspberry Pi 4 / Pi 5:** Runs AI algorithms for face and object detection.
- **Camera Module:** Captures video/images for detection tasks.

Mini Drone (PAS)

- **Cinelog 35 Frame:** Compact FPV drone for tight spaces.
- **SpeedyBee F7 Stack:** Controls motors and onboard electronics.
- **FPV Camera & VTX:** Provides live video feed to pilot.
- **FPV Goggles & Controller:** Interface for manual piloting of PAS.

Software:

Low-Level / Microcontroller

- **C/C++/Python**:
  - Write custom flight control code.
  - Control motors, sensors, and payload release mechanisms on Arduino/ Pixhawk.

Libraries

- **OpenCV**: Image/video processing, face detection, preprocessing.
- **YOLOv8**: Object detection (faces, victims, other objects).
- **Insightface**: Face identification & verification.
- **TensorFlow**: ML platform for building and training models (used for YOLOv8, FaceNet, etc.).

. Uvicorn
Lightweight ASGI server for fast Python web applications.

. NumPy
Library for numerical computing and multidimensional array operations.

. Pydantic
Data validation library using Python type hints.

. Pillow
Python library for image processing and editing.

. SciPy
Scientific computing library with advanced mathematical tools.

. PyMongo
Python driver for MongoDB database connectivity.

. pyttsx3
Offline text-to-speech library for Python applications.

Backend

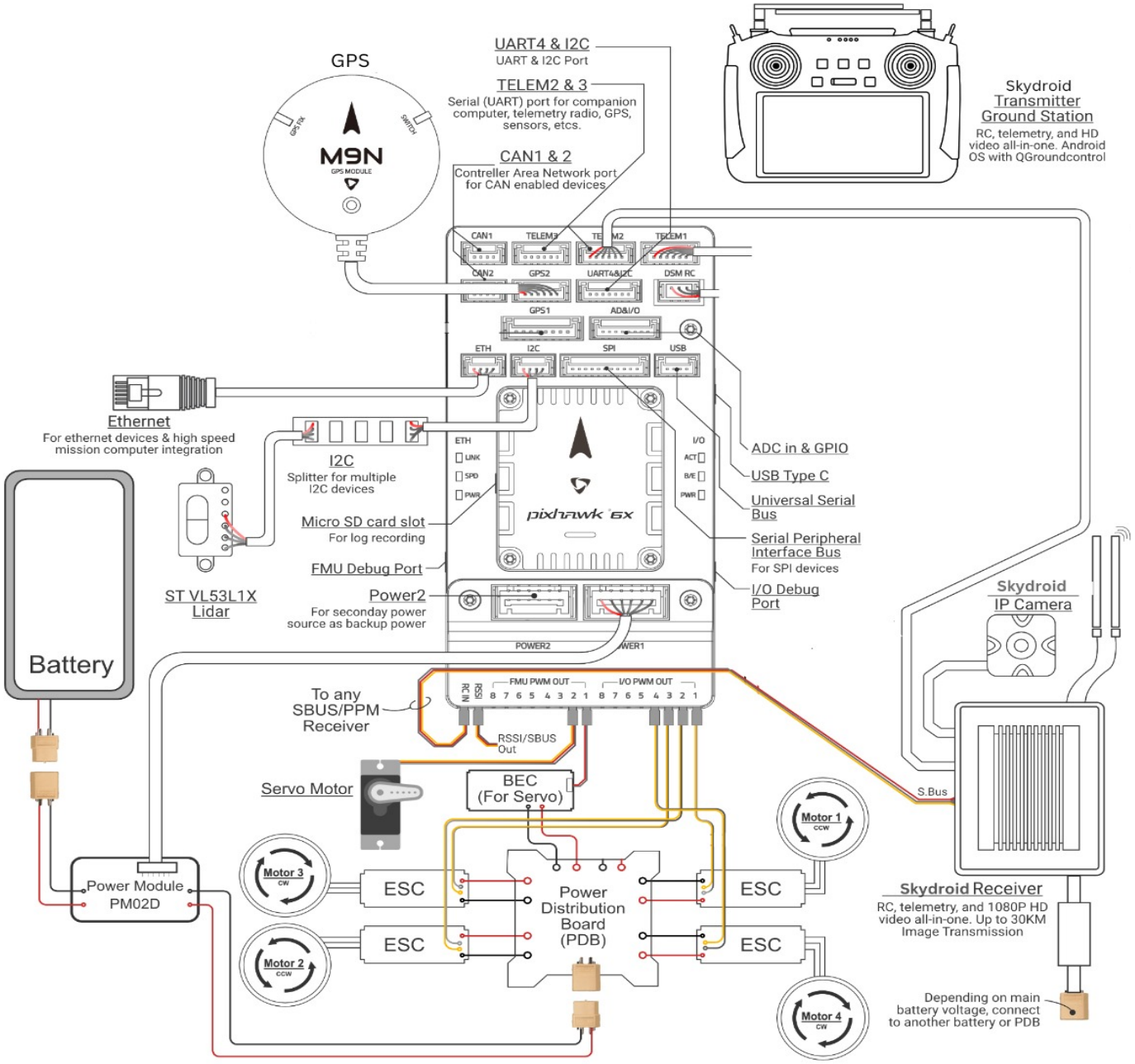
- **Python**: Main programming language for AI modules, drone control scripts, and communication.
- **FastAPI**: Lightweight backend APIs for mission coordination, face detection alerts, or drone commands.

Database

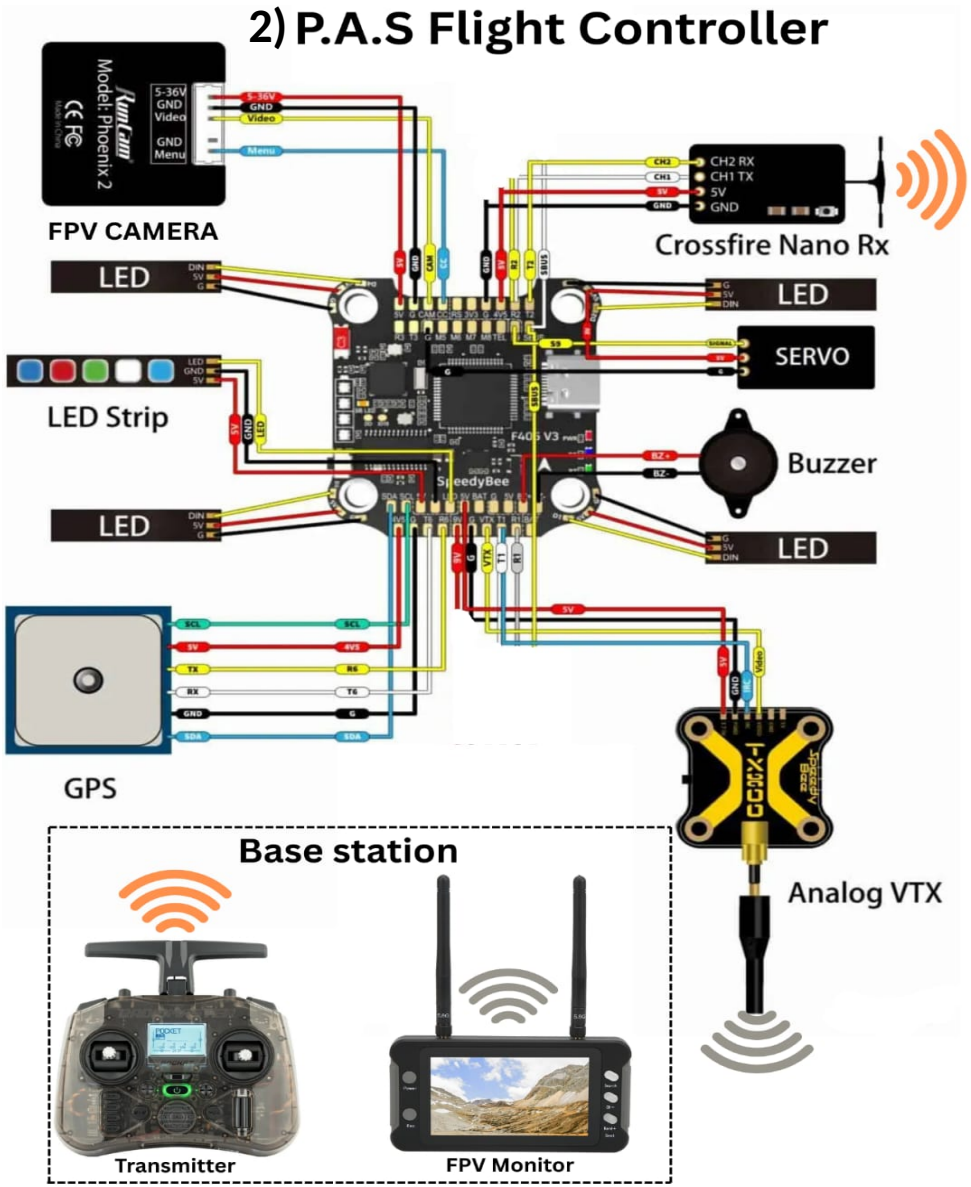
- **MongoDB**: MongoDB is a NoSQL document-oriented database providing flexible schema, scalable storage, high availability, indexing, and efficient real-time data management support.

Circuit Diagram & PCB Layout:

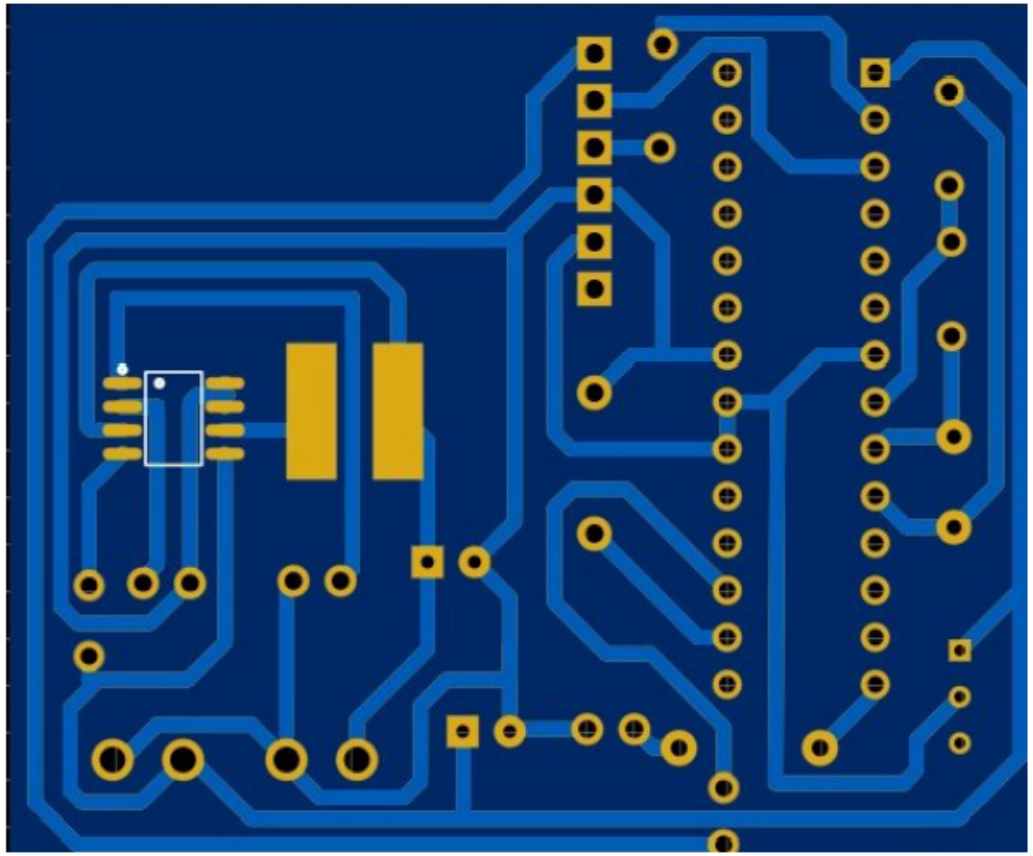
1)M.A.S:



Circuit Diagram :



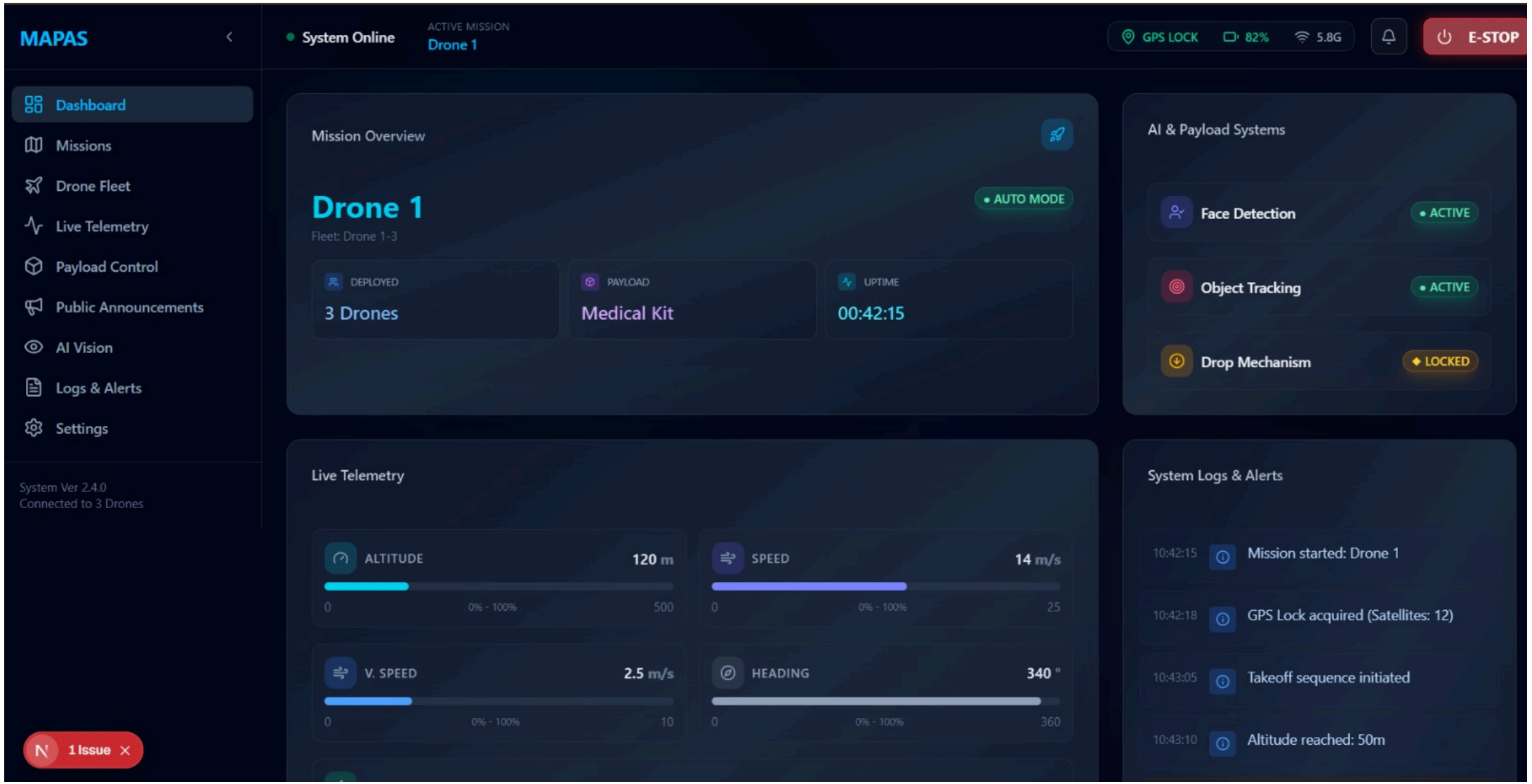
PCB Layout:



Bill of Materials:

SL No.	Item	Qty	Unit Price	Amount
1	Tarot Foldable Hexa-copter Frame TL810A	1	17,469.00	17,469.00
2	Cube Orange Pro flight controller	1	67,654.00	67,654.00
3	skydroid T12 transmitter	1	22,000.00	22,000.00
4	Tattu 22.2v 25c 6s 22000mah lipo battery	2	26,987.00	53,974.00
5	T Motor NS16x5.4 Propeller	4	2,763.00	11,052.00
6	Reflex Drive Q-Series ESC 6S 80A	6	2,497.00	14,982.00
7	T motor antigravity mn5008 kv340 6s	6	9,843.00	59,058.00
8	SKYRC PC1080 Dual Channel Charger	1	28,753.00	28,753.00
9	NVIDIA Jetson Orin Nano 8GB Module	1	32,753.00	32,753.00
10	Mini Carrier Board for Jetson Orin™ Nano/Orin™ NX	1	17,543.00	17,543.00
11	20MP (5K) AR2020 High Resolution Camera for NVIDIA®	1	27,894.00	27,894.00
12	Miscellaneous parts	1	25,000.00	25,000.00

Modelling & Simulation of UI Design:



Conclusion and Future Scope:

Conclusion:

- MAPAS offers a smart dual-drone system that combines autonomous flight with manual precision, making it highly versatile for different missions.

- The modular payload design allows it to adapt quickly to tasks like rescue, aid delivery, agriculture, and surveillance.

- It improves response time in emergencies and increases access to hard-to-reach or dangerous areas without risking human lives.

- Supports multiple sectors, including disaster management, farming, public safety, and environmental monitoring.

- We have successfully completed about 90% of the Mother Drone (MAS), including most of the hardware setup and flight control system. Around 40% of the software part is also done, covering the basic functions like controls and telemetry.

- We have also planned the sourcing of parts for the PAS (small drone) so that its assembly can start soon.

In the upcoming phase, we aim to finalize software integration, complete field testing, and optimize communication between MAS and PAS for real-time coordinated missions.

Future Scope:

1. AI-Based Smart Decision Making

The system can be upgraded with advanced AI and machine learning algorithms for automatic object detection, disaster assessment, crop analysis, and intelligent mission planning, enabling fully autonomous and faster decision-making.

2. Swarm Drone Technology

Multiple drones can be deployed in coordinated swarm formations to cover larger areas efficiently for disaster rescue, agricultural monitoring, surveillance, and emergency response operations.

3. Autonomous Delivery System Expansion

The drone system can be expanded for large-scale medical logistics, including emergency blood transport, vaccine delivery, organ transport, and rapid supply distribution in remote or disaster-affected areas.

4. Hybrid Autonomous and Piloted Mode Optimization

Further improvement in seamless switching between autonomous and manual (piloted) modes can enhance mission flexibility, safety, and operational control in complex real-world environments.

5. Lightweight High-Capacity Payload Design

Future research can focus on advanced lightweight composite materials and improved structural design to increase payload capacity without compromising flight efficiency or battery life.

References:

[1] S.D. Panjaitan et al. “***Drone Technology Implementation Approach to Conventional Paddy Fields Application"*** International Research Journal of Modernization in Engineering, Technology and Science, vol. 5, no. 5, pp. 5259–5268, May 2023.

[2] A. Saha et al. ***"FPV Drone with GPS used for Surveillance in Remote Areas"*** International Research Journal of Modernization in Engineering, Technology and Science, vol. 5, no. 10, pp. 3056–3059, Oct. 2023.

[3] D. He et al. ***"Drone-Assisted Public Safety Networks: The Security Aspect,"*** Journal of Software Engineering and Applications, vol. 16, no. 6, pp. 223–263, Jun. 2023.

[4] G. Singh et al. ***"Drone Food Delivery: A Solution to Crowding During the Global COVID-19 Pandemic,"*** 2025 International Conference on Computing and Communication Technologies (ICCCT), Chennai, India, 2025, pp. 1–8.

[5] Gang Xiang et al.***"Design of the Life-ring Drone Delivery System for Rip Current Rescue,"*** 2022 IEEE 48th Annual Conference of the Industrial Electronics Society (IECON), Brussels, Belgium, 2022, pp. 1–8.